

PostalMS

Team Meeting Minutes

5 Meetings | 3 March 2026 - 14 April 2026

Meeting 1

Date	Monday 3 March 2026
Time	2:00pm - 3:00pm
Location	Ritterman Building, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyurbhai Patel, Forhad Hossain
Absent	None

Agenda

- Discuss the project requirements from the CST2550 brief
- Agree on roles and responsibilities for each team member
- Set up the GitHub repository structure
- Plan the first sprint tasks and deadlines

Discussion

- The team met in the Ritterman Building to go through the CST2550 coursework brief. The project was confirmed as a Postal Service Management System using C# WinForms and Microsoft SQL Server. Three data structures were identified: Hash Table for $O(1)$ parcel lookup, BST for $O(\log n)$ sorted display and Queue for $O(1)$ FIFO delivery processing.

Decisions Made

- Project name confirmed as PostalMS
- Application type confirmed as C# WinForms desktop application
- Database confirmed as Microsoft SQL Server 2022
- Three custom data structures selected: Hash Table, BST, Queue
- GitHub repository to be created by the Team Leader
- Roles: Durga (Team Leader + Developer), Meet (Secretary), Tapan (Developer), Forhad (Tester)

Action Points

Assigned To	Action
Durga	Create GitHub repository and push initial C# WinForms project by 6 March
Meet	Set up meetings folder and begin documenting meeting minutes
Tapan	Research BST recursive implementation approach
Forhad	Research MSTest unit testing framework and review coursework brief

Next Meeting

- Monday 10 March 2026 -- Sheppard Library, Middlesex University

Meeting 2

Date	Monday 10 March 2026
Time	1:00pm - 2:30pm
Location	Sheppard Library, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyurbhai Patel, Forhad Hossain
Absent	None

Agenda

- Review progress from Sprint 1
- Review the GitHub repository setup
- Discuss and agree on the database schema design
- Discuss data structure pseudocode and time complexity
- Plan Sprint 2 tasks

Discussion

- The team reviewed Sprint 1 progress. Durga presented the GitHub repository with the C# WinForms project set up. The database schema was agreed with three tables: Users, Parcel and Delivery. The recursive BST approach was agreed and the Hash Table chaining design was confirmed with load factor 0.75.

Decisions Made

- Database to have three tables: Users, Parcel and Delivery
- Hash Table to use chaining for collision resolution with load factor 0.75
- BST to use recursive insert, search and delete operations
- Queue to be implemented as a linked list with front and rear pointers
- Class diagram and ERD to be created in draw.io

Action Points

Assigned To	Action
Durga	Implement CustomHashTable and CustomBST classes from scratch
Durga	Write the complete SQL database script with all tables and stored procedures
Tapan	Implement CustomQueue class from scratch
Meet	Create class diagram and ERD in draw.io
Forhad	Begin writing unit tests for Hash Table and Queue

Next Meeting

- Monday 17 March 2026 -- Ritterman Building, Middlesex University

Meeting 3

Date	Monday 17 March 2026
Time	2:00pm - 3:30pm
Location	Ritterman Building, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyurbhai Patel, Forhad Hossain
Absent	None

Agenda

- Review Sprint 2 progress and data structure implementations
- Discuss UI design and application navigation structure
- Review the database script and sample data
- Review initial unit test cases
- Plan Sprint 3 tasks

Discussion

- Durga demonstrated the working CustomHashTable, CustomBST and CustomQueue implementations. The SQL database script was reviewed by the full team. The UI design was discussed -- the red colour theme was agreed for the entire application. Navigation was confirmed as Home, Parcels, Deliveries, Help and Info. An offline AI assistant was proposed as a bonus feature.

Decisions Made

- Red colour theme confirmed for the entire PostalMS application
- Navigation confirmed: Home, Parcels, Deliveries, Help and Info pages
- AI assistant to be implemented as offline rule-based system requiring no internet
- Unit tests to cover all three data structures, pricing logic and tracking ID generation
- Pseudocode document to be written with full explanations for all three data structures

Action Points

Assigned To	Action
Durga	Implement all WinForms pages and connect to SQL database
Durga	Implement offline AI assistant with parcel tracking and price estimation
Tapan	Assist with WinForms layouts and record MP4 video
Meet	Write complete pseudocode document and begin PDF report
Forhad	Complete all 59-unit tests and confirm all pass

Next Meeting

- Monday 24 March 2026 -- Sheppard Library, Middlesex University

Meeting 4

Date	Monday 24 March 2026
Time	1:00pm - 3:00pm
Location	Sheppard Library, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyurbhai Patel, Forhad Hossain
Absent	None

Agenda

- Final review of all completed features and source code
- Review unit test results and confirm all 59 tests pass
- Review report and pseudocode document progress
- Plan the video demonstration
- Confirm final submission checklist

Discussion

- Durga demonstrated the complete PostalMS application. Forhad ran all 59 unit tests live -- all passed in 66ms with 0 failures covering Hash Table (12), BST (13), Queue (10), price calculation (9), tracking ID (6) and integration (5). Meet presented the pseudocode document. The video plan was agreed. The submission checklist was reviewed.

Decisions Made

- All source code to be pushed to GitHub by 7 April 2026
- PDF report to be completed by 8 April 2026
- Video to be recorded and uploaded by 9 April 2026
- Submission .txt file before 10 April 2026 at 5:30pm
- All 4 markers to be invited as collaborators

Action Points

Assigned To	Action
Durga	Final code review, push all files to GitHub, invite all 4 markers
Meet	Complete and upload the final PDF report
Tapan	Record the MP4 video demonstration
Forhad	Upload unit testing report and check all repo folders

Next Meeting

- . Monday 14 April 2026 -- 2:00pm - 4:00pm, Ritterman Building, Middlesex University

Meeting 5 (Sprint 4 -- Feature Enhancements)

Date	Monday 14 April 2026
Time	2:00pm - 4:00pm
Location	Ritterman Building, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyurbhai Patel, Forhad Hossain
Absent	None

Agenda

- Review professor feedback and identify required changes
- Plan and prioritise all Sprint 4 feature improvements
- Assign Sprint 4 tasks to team members
- Review stamps ordering system design
- Review Find Us city detection design
- Confirm 78 unit test plan
- Plan document updates for all supporting documents

Discussion

- The team met to review professor feedback and plan Sprint 4. The following areas were identified for improvement: stamps section, Find Us page, receiver details, email validation, ProfileView and drop off locations.
- Durga presented the StampsView plan. The stamps section was to be completely removed from ParcelsView and rebuilt as a standalone page with three tabs: Browse Stamps, Order Stamps and My Stamp Orders. The ordering flow was designed like a food delivery app with two clickable cards for Delivery or Click and Collect. Delivery requires name, full address, city, postcode and email. Click and Collect requires name, phone number, date of birth, location, date, time slot and email. Payment offers Pay Now Online or Pay In Store -- Pay In Store is only available with collection and shows a validation message otherwise.
- Missed pickup logic was agreed: if a slot is missed the order is held for 48 hours then auto-rescheduled to the next available slot at the same location. A red banner appears in My Stamp Orders to notify the user. The price breakdown shows stamps subtotal, delivery fee (Standard GBP 1.50, Express GBP 3.99, Collect Free), 20% VAT and 2% service fee.
- The FindUsView redesign was agreed. Seven UK cities were confirmed: London (12 locations), Manchester (12), Birmingham (12), Leeds (10), Bristol (10), Edinburgh (10) and Glasgow (10) -- 76 locations total. The page auto-detects the user city from the City column in the Users database table. A city dropdown allows browsing other cities. Each location card shows name, address, phone, opening hours, travel directions and a Google Maps link.
- ParcelsView improvements were agreed. Step 4 receiver details to be expanded to full name, phone number, email, street address, city, postcode and special delivery instructions. A UK city dropdown to be added for domestic destinations. The international country dropdown overlap with Step 4 was to be fixed. Drop off locations already existed but were confirmed to keep Google Maps links.
- ProfileView was reported as showing a blank page for some users due to a missing column query. Durga confirmed the fix using a direct Q_Public query for columns that definitely exist in the database.

- Gmail validation was confirmed to be enforced across all four forms: LoginForm, RegisterForm, StampsView and ParcelsView. This was a professor requirement.
- Forhad presented the 78 test plan. Three new test classes would be added: GmailValidationTests (8 tests), StampPriceTests (6 tests) and CityLocationTests (8 tests). The test file would be made fully self-contained with local copies of all data structures to avoid project reference errors.
- Meet confirmed all four supporting documents would be updated: Unit Testing Report (78 tests, 9 classes), Sprint Planning (Sprint 4 with 26 tasks), Pseudocode (Gmail, stamps, city, receiver algorithms), Time Complexity (Sprint 4 algorithms).

Decisions Made

- Stamps section removed from ParcelsView and rebuilt as standalone StampsView
- Stamp ordering to use food-app style Delivery vs Collect card selection
- Delivery details: name, address, city, postcode, email, delivery speed
- Collect details: name, phone number, date of birth, location, date, time slot, email
- Pay In Store available only with Click and Collect -- validation message otherwise
- Missed pickup: 48 hour hold then auto-reschedule with red banner notification
- Stamp price: stamps + delivery + 20% VAT + 2% service fee
- FindUsView: 7 cities, 76 locations total, auto-detect city from user profile
- City field added to RegisterForm and Users database table
- RegisterUser stored procedure updated to accept and store city
- ParcelsView Step 4 expanded to full receiver details with all fields
- UK city dropdown added for domestic parcel destination
- International dropdown overlap with Step 4 fixed
- ProfileView blank page bug fixed using Q_Public direct query
- Gmail validation enforced in all four forms across the application
- 78 unit tests: 59 original plus 19 new across 3 new test classes
- Test file made self-contained with no external project reference
- All 4 supporting documents updated to reflect Sprint 4 changes

Action Points

Assigned To	Action
Durga	Remove stamps tab from ParcelsView and build complete StampsView from scratch
Durga	Implement Delivery vs Collect cards and Pay Now vs Pay In Store cards in StampsView
Durga	Add full delivery details (name, address, city, postcode, email, speed) to stamp ordering
Durga	Add full collect details (name, phone, DOB, location, date, slot, email) to stamp ordering
Durga	Implement missed pickup logic: 48 hour hold, auto-reschedule and red banner
Durga	Expand ParcelsView Step 4 with full receiver details including all fields
Durga	Add UK city dropdown and fix international dropdown overlap in ParcelsView
Durga	Rewrite FindUsView with 7 cities, 76 locations and city auto-detection

Assigned To	Action
Durga	Add City field to RegisterForm and update database table and stored procedure
Durga	Fix ProfileView blank page bug using Q_Public direct query
Durga	Enforce Gmail validation in all four forms across the application
Forhad	Write 19 new unit tests across GmailValidationTests, StampPriceTests and CityLocationTests
Forhad	Make test file self-contained and confirm all 78 tests pass with 0 failures
Meet	Update Unit Testing Report to document all 78 tests across 9 classes
Meet	Update Sprint Planning with Sprint 4 containing all 26 tasks
Meet	Update Pseudocode with Gmail, stamp price, city detection and receiver algorithms
Meet	Update Time Complexity with Sprint 4 algorithm analysis
Tapan	Assist with end-to-end testing of all new Sprint 4 features
Durga	Push all Sprint 4 changes to GitHub after all features confirmed working

Next Meeting

- No further meetings scheduled. All Sprint 4 tasks to be completed and pushed to GitHub by Friday 19 April 2026.

Meetings Summary

No.	Date	Duration	Location	Key Outcome
1	Monday 3 March 2026	1 hour	Ritterman Building	Roles assigned, data structures chosen, repo planned
2	Monday 10 March 2026	1.5 hours	Sheppard Library	Database schema agreed, data structures designed
3	Monday 17 March 2026	1.5 hours	Ritterman Building	Red theme confirmed, navigation agreed, AI planned
4	Monday 24 March 2026	2 hours	Sheppard Library	59 tests all passing, final submission plan confirmed
5	Monday 14 April 2026	2 hours	Ritterman Building	Sprint 4 planned, 78 tests, all 26 tasks assigned